SUSPENSION GEOMETRY — HARDPOINTS 2027

Double A-arm (SLA), front and rear | FSAE 2027 -- PUCPR Racing

Sign conventions: negative camber = top of the wheel inboard; toe-in positive; positive force = tension. Lengths in mm, angles in degrees unless stated. Two coordinate frames appear in this document and every table names the one it uses — see Provenance below.

Generated 2026-08-25 by scripts/build_summary_doc.py from sla_geometry.py and steering_geometry.py. Do not hand-edit: change the config in those scripts and re-run. This document supersedes "Hardpoints Suspensão 2027", whose rate tables predated the upright-rotation sign fix (commit 0e7e524) and whose force tables assumed a different weight distribution from its own roll-stiffness section.

0. Provenance and frames

Content	Produced by
Wishbone hardpoints, static KPIs, rates, roll, anti-geometry, leg forces	sla_geometry.py
Caster, outboard ball joints, tie rod, rack, bump steer, Ackermann, effort	steering_geometry.py
3D kinematic solve behind the steering numbers	vdcore.geometry.solver.DWSolver
Composition, cross-checks, this document	scripts/geometry_summary.py + build_summary_doc.py

Frame	Axes	Used in
DESIGN	y outboard, z up, x REARWARD	Sections 2 and 3 (front-view work)
ISO 8855	X FORWARD, Y LEFT, Z up	Section 6 (model export)

Conversion: X_iso = −x_rearward, Y_iso = ±y_outboard, Z_iso = z. In ISO 8855 the rear axle is at NEGATIVE X and right-hand corners are at NEGATIVE Y.

OPEN: the rear tie rod is not synthesised by any script. steering_geometry.py covers the front axle only, so RL/RR TIE_ROD_IN and TIE_ROD_OUT are hand-entered points read from legacy_app/carro_formula_2027.csv. They are not regenerated, not bounds-checked and not covered by any test.

1. Vehicle, stiffness and rules compliance

Parameter	Value
Total / sprung mass	315.0 / 270.0 kg
Wheelbase	1540.0 mm
Front track / rear track	1240.0 / 1200.0 mm
CG height / station	320.0 mm / 693.0 mm aft of the front axle
Static front mass fraction	55.0 %
Roll axis height at the CG	44.0 mm
Roll moment arm	276.0 mm
Target roll gradient	1.00 °/g
Required roll stiffness	731.0 N·m/°
Chassis torsional stiffness	2193 (min) / 3655 (target) N·m/°

The front mass fraction sets BOTH the roll axis height and the leg-force load cases in section 7. It is a single input; the two must never be allowed to disagree again.

Status	Item	Value	Target
OK	Wheelbase	1540 mm	1525 to 10000
OK	Narrow / wide track ratio	96.8 %	75 to 100
OK	60° tilt test	1200 mm fitted	needs 1109 mm

The tilt test uses the NARROWER track, which is the rear (1200 mm). Quoting that as "the track" hides the 1240 mm on the other axle.

2. Front suspension

Front-view points

DESIGN frame: y positive OUTBOARD, z positive up, origin at ground level on the car centreline. Two decimals are required — the integers printed in the superseded document could not reproduce its own FVIC or roll centre.

Point	y [mm]	z [mm]
Lower ball joint (LBJ)	582.00	130.00
Upper ball joint (UBJ)	536.97	385.40
LCA inboard	175.00	117.38
UCA inboard	175.00	308.58
FVIC (construction)	-880.00	84.68

Wishbones

Parameter	Value
LCA / UCA length (front view)	407.20 / 370.03 mm (ratio 0.909)
Outboard vertical separation	255.40 mm
Inboard vertical separation	191.20 mm
LCA inclination	1.78° (falls from wheel to chassis)
UCA inclination	11.98° (falls from wheel to chassis)
RC that would flatten the LCA	53.73 mm

Checks

Status	Item	Value	Target
OK	Roll centre height	35.00 mm	20 to 70
OK	FVSA length	1500.00 mm	1300 to 1700
OK	Scrub radius	15.08 mm	5 to 25
OK	KPI	10.00 °	6 to 14
OK	Kingpin length (front view)	259.34 mm	200 to 260
OK	LCA length (front view)	407.20 mm	320 to 430
OK	UCA / LCA ratio	0.909	0.55 to 0.98
OK	Camber gain	0.0382 °/mm	0.03 to 0.05
OK	FVIC on the far side of the car	-880.00 mm	–
OK	Ball joints inside the rim	130 / 385 mm	80 to 410

Kingpin and arm lengths here are FRONT-VIEW projections — the right quantity for the swing-arm construction, but not the length that gets cut. Section 5 checks the true 3D members.

Rates about static

Quantity	Value
Camber gain	-0.0382 °/mm
Roll centre migration	-0.3873 mm/mm
Half-track change	+0.0565 mm/mm
Camber at full bump	-2.47°
Camber at full droop	-0.56°
RC over the travel range	25.4 to 44.8 mm

Travel ±25 mm. Camber gain equals 57.2958 / FVSA = 0.0382 °/mm, the textbook value for a wheel turning about an instant centre 1500 mm away. Half-track change is POSITIVE: the contact patch moves outboard in bump.

At 1.5° of roll

Quantity	Value
Outer wheel camber vs road	-0.63° (static -1.50°)
Inner wheel camber vs road	-2.39°
Roll centre height	33.9 mm (design 35.0)
Roll centre lateral migration	-110.2 mm
Wheel travel at that roll	16.23 mm
Outer camber in the useful window	OK (-2.5 to 0.0°)

Longitudinal layout and anti-geometry

DESIGN frame, x positive REARWARD. Sweep and e/a as BUILT are in section 5, measured off the caster-corrected ball joints.

Parameter	Value
LCA pickups x	-130.0 / 130.0 mm
UCA pickups x	-120.0 / 120.0 mm
LCA / UCA e/a ratio	see section 5 — measured as built (target ≤ 1.5)
Anti-dive	0.00 % (target 0 to 30)

All pivot axes are exactly horizontal, so the side-view instant centre is at infinity and the anti-feature is identically zero — not approximately zero. That is a design decision to defend, not a check that happened to pass.

3. Rear suspension

Front-view points

DESIGN frame: y positive OUTBOARD, z positive up, origin at ground level on the car centreline. Two decimals are required — the integers printed in the superseded document could not reproduce its own FVIC or roll centre.

Point	y [mm]	z [mm]
Lower ball joint (LBJ)	558.60	130.00
Upper ball joint (UBJ)	519.69	390.34

LCA inboard	175.00	129.53
UCA inboard	175.00	321.91
FVIC (construction)	-800.00	128.33

Wishbones

Parameter	Value
LCA / UCA length (front view)	383.60 / 351.42 mm (ratio 0.916)
Outboard vertical separation	260.34 mm
Inboard vertical separation	192.38 mm
LCA inclination	0.07° (falls from wheel to chassis)
UCA inclination	11.23° (falls from wheel to chassis)
RC that would flatten the LCA	55.71 mm

Checks

Status	Item	Value	Target
OK	Roll centre height	55.00 mm	20 to 70
OK	FVSA length	1400.00 mm	1300 to 1700
OK	Scrub radius	21.97 mm	5 to 25
OK	KPI	8.50 °	3 to 10
FAIL	Kingpin length (front view)	263.23 mm	200 to 260
OK	LCA length (front view)	383.60 mm	320 to 430
OK	UCA / LCA ratio	0.916	0.55 to 0.98
OK	Camber gain	0.0409 °/mm	0.03 to 0.05
OK	FVIC on the far side of the car	-800.00 mm	–
OK	Ball joints inside the rim	130 / 390 mm	80 to 410

Kingpin and arm lengths here are FRONT-VIEW projections — the right quantity for the swing-arm construction, but not the length that gets cut. Section 5 checks the true 3D members.

Rates about static

Quantity	Value
Camber gain	-0.0409 °/mm
Roll centre migration	-0.4193 mm/mm
Half-track change	+0.0917 mm/mm
Camber at full bump	-2.54°
Camber at full droop	-0.49°
RC over the travel range	44.7 to 65.7 mm

Travel ±25 mm. Camber gain equals 57.2958 / FVSA = 0.0409 °/mm, the textbook value for a wheel turning about an instant centre 1400 mm away. Half-track change is POSITIVE: the contact patch moves outboard in bump.

At 1.5° of roll

Quantity	Value
Outer wheel camber vs road	-0.65° (static -1.50°)
Inner wheel camber vs road	-2.36°
Roll centre height	54.3 mm (design 55.0)
Roll centre lateral migration	-70.3 mm
Wheel travel at that roll	15.70 mm
Outer camber in the useful window	OK (-2.5 to 0.0°)

Longitudinal layout and anti-geometry

DESIGN frame, x positive REARWARD. Sweep and e/a as BUILT are in section 5, measured off the caster-corrected ball joints.

Parameter	Value
LCA pickups x	1140.0 / 1480.0 mm
UCA pickups x	1160.0 / 1480.0 mm
LCA / UCA e/a ratio	see section 5 — measured as built (target ≤ 1.5)
Anti-squat	0.00 % (target 0 to 30)

All pivot axes are exactly horizontal, so the side-view instant centre is at infinity and the anti-feature is identically zero — not approximately zero. That is a design decision to defend, not a check that happened to pass.

4. Steering (front axle)

Absent in its entirety from the superseded document. Caster and mechanical trail set steering feel and effort; bump steer is the tie rod's primary KPI.

Parameter	Value
Caster angle	5.00°
Mechanical trail	21.43 mm
Scrub radius (3D)	15.08 mm
Steering arm length	80.00 mm
Tie rod length	340.79 mm
Rack position (x rearward, z)	30.0, 158.3 mm
Rack half length / max travel	270.0 / ±38.0 mm

Quantity	Value
Bump steer (per side)	-0.00011 °/mm
Bump steer (total toe)	+0.00021 °/mm
Toe at full bump / droop (per side)	+0.171 / +0.162 °
C-factor	-1.278 mm/°
Steering ratio	4.58 : 1
Max steer at stroke (outer / inner)	27.00 / 38.07°
Ackermann at target steer	66.8 %
Worst rod-end misalignment	9.78°

Bump steer is per side and total toe separately, because per-side versus total has caused confusion on the real car. The gradient is effectively zero at static: both tie rods aim at their own front-view instant centre. The residual is second order and the same sign at both ends of travel, so roll steer stays near zero while total toe rises slightly at full travel.

Steering effort (parking)

Quantity	Value
Fz per front wheel	849.8 N
Kingpin moment per wheel	22.27 N·m
Rack force (both wheels)	608.4 N
Steering wheel torque	9.73 N·m
Rim force	69.5 N

Parking effort assumes a friction coefficient. PUCPR Racing has no TTC tyre data — this is a design_intent assumption, not a measured value.

5. Member lengths and sweep as built (true 3D)

Measured off the merged hardpoints in ISO 8855, after caster. These are the members that get cut, as opposed to the front-view projections checked in sections 2 and 3.

Front axle [FL]

Status	Item	Value	Target
OK	LCA front leg	424.49 mm	320 to 430
FAIL	LCA rear leg	430.61 mm	320 to 430
OK	UCA front leg	392.96 mm	320 to 430
OK	UCA rear leg	385.39 mm	320 to 430
OK	Kingpin length, front view	259.34 mm	200 to 260
FAIL	Kingpin length, TRUE 3D	260.30 mm	200 to 260

Parameter	Value
LCA base / sweep as built	260.00 / 10.06 mm (e/a 0.077)
UCA base / sweep as built	240.00 / 12.28 mm (e/a 0.102)
Tie rod	340.79 mm

Rear axle [RL]

Status	Item	Value	Target
FAIL	LCA front leg	554.21 mm	320 to 430
OK	LCA rear leg	388.26 mm	320 to 430
FAIL	UCA front leg	517.59 mm	320 to 430
OK	UCA rear leg	356.50 mm	320 to 430
FAIL	Kingpin length, front view	263.23 mm	200 to 260
FAIL	Kingpin length, TRUE 3D	263.23 mm	200 to 260

Parameter	Value

LCA base / sweep as built	340.00 / 230.00 mm (e/a 1.353)
UCA base / sweep as built	320.00 / 220.00 mm (e/a 1.375)
Tie rod	364.59 mm

Any leg flagged above is longer than the packaging window the front-view check passes it against. The swept rear legs matter most: e/a above 1 means the sweep roughly doubles the leg load, and buckling of a long compression member is outside this tool's scope. Someone has to check it.

6. Merged hardpoints (ISO 8855)

X POSITIVE FORWARD. Y POSITIVE LEFT. Z POSITIVE UP. Origin at the front axle centreline, ground plane, vehicle centreline. The rear axle is therefore at NEGATIVE X and right-hand corners at NEGATIVE Y. This is NOT the design frame used in sections 2 and 3.

FL

Point	X [mm]	Y [mm]	Z [mm]
UCA_IN_FRONT	120.00	175.00	308.58
UCA_IN_REAR	-120.00	175.00	308.58
UCA_OUT	-12.28	536.97	385.40
LCA_IN_FRONT	130.00	175.00	117.38
LCA_IN_REAR	-130.00	175.00	117.38
LCA_OUT	10.06	582.00	130.00
TIE_ROD_IN	-30.00	270.00	158.30
TIE_ROD_OUT	84.59	590.29	178.75
WHEEL_CENTER *	0.00	620.00	245.00
CONTACT_PATCH *	0.00	620.00	0.00

FR

Point	X [mm]	Y [mm]	Z [mm]
UCA_IN_FRONT	120.00	-175.00	308.58
UCA_IN_REAR	-120.00	-175.00	308.58
UCA_OUT	-12.28	-536.97	385.40
LCA_IN_FRONT	130.00	-175.00	117.38
LCA_IN_REAR	-130.00	-175.00	117.38
LCA_OUT	10.06	-582.00	130.00
TIE_ROD_IN	-30.00	-270.00	158.30
TIE_ROD_OUT	84.59	-590.29	178.75
WHEEL_CENTER *	0.00	-620.00	245.00
CONTACT_PATCH *	0.00	-620.00	0.00

RL

Point	X [mm]	Y [mm]	Z [mm]
UCA_IN_FRONT	-1160.00	175.00	321.91

UCA_IN_REAR	-1480.00	175.00	321.91
UCA_OUT	-1540.00	519.69	390.34
LCA_IN_FRONT	-1140.00	175.00	129.53
LCA_IN_REAR	-1480.00	175.00	129.53
LCA_OUT	-1540.00	558.60	130.00
TIE_ROD_IN	-1480.00	175.00	169.00
TIE_ROD_OUT	-1594.60	520.81	183.53
WHEEL_CENTER *	-1540.00	600.00	245.00
CONTACT_PATCH *	-1540.00	600.00	0.00

RR

Point	X [mm]	Y [mm]	Z [mm]
UCA_IN_FRONT	-1160.00	-175.00	321.91
UCA_IN_REAR	-1480.00	-175.00	321.91
UCA_OUT	-1540.00	-519.69	390.34
LCA_IN_FRONT	-1140.00	-175.00	129.53
LCA_IN_REAR	-1480.00	-175.00	129.53
LCA_OUT	-1540.00	-558.60	130.00
TIE_ROD_IN	-1480.00	-175.00	169.00
TIE_ROD_OUT	-1594.60	-520.81	183.53
WHEEL_CENTER *	-1540.00	-600.00	245.00
CONTACT_PATCH *	-1540.00	-600.00	0.00

** reference points, not hardpoints.*

STATIC ALIGNMENT ENCODED HERE: camber +0.00°, toe +0.00° per side. CONTACT_PATCH sits directly below WHEEL_CENTER on all four corners, which is a zero-camber, zero-toe convention, while the rate and roll tables assume the design-intent static camber. Decide the convention once and apply it everywhere — see section 8.

7. Load case behind the leg forces

The superseded document printed leg forces with no stated load case, so they could not be checked or defended. These are the assumptions.

Assumption	Value	Source
Total mass	315.0 kg	input
Front mass fraction	55.0 %	input
CG height	320.0 mm	estimate
Lateral friction coefficient	1.50	design_intent
Longitudinal friction coefficient	1.40	design_intent
Lateral acceleration	1.50 g	design_intent
Lateral load transfer distribution	0.55	design_intent

TWO LIMITS, to be repeated wherever the force table appears:

1. The friction coefficients are a TYRE assumption. PUCPR Racing has no TTC data, so every force inherits an unmeasured input.

2. There is no pushrod in this model. An upright on two A-arms and a tie rod has five constraints against six degrees of freedom; the sixth member is the pushrod. The solve closes the system by treating the UCA as a two-force member with no X component and enforcing only the moment about the roll axis, and it places the front ball joints at zero caster, so braking generates no kingpin moment and the tie rod carries nothing.

These are screening numbers. They are NOT a load set to size tubes from.

8. Open items

Item	Effect	Consequence
Front scrub if static camber were built in	15.08 → 21.49 mm	still in window
Rear scrub if static camber were built in	21.97 → 28.39 mm	leaves the 5 to 25 window

The static camber / toe convention has to be decided once: either the export carries it and scrub radius moves, or it does not and the rate tables stop claiming it.

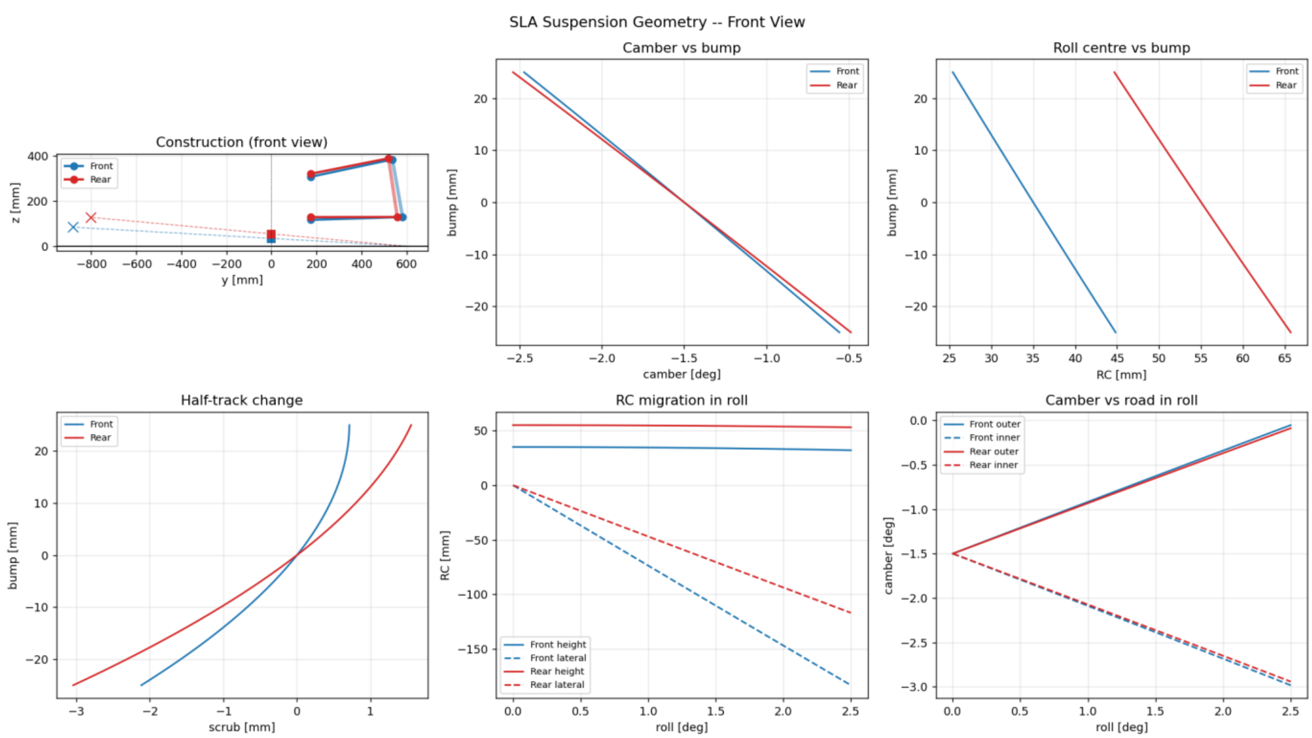
The front mass fraction is an unusual value for a rear-drive car and it drives both the roll-stiffness requirement and every leg force. Confirm it before the chassis is sized against it.

The rear tie rod has no synthesis script and no test.

Deliberately not computed here, and out of scope for this tool: wheel rates, motion ratio, ride frequency, damping; stress, deflection, fatigue, buckling; anything requiring tyre data until TTC data is acquired.

9. Charts

Regenerated from sla_geometry.py in its corrected mode. The plots in the superseded document came from a mix of toolchains and disagreed with its own tables.



Front-view construction, camber and roll centre vs travel, half-track change, and roll behaviour — front and rear.